

Pandas

columns

rows

name	age		

heterogeneity

column name

json = { }

csv

xlsx

tsv

Pandas

- ⇒ Open source library
- ⇒ easy to use for data analysis
- ⇒ high performance

Application

Data analysis

Data Science

Modelling

visualization

Text processing

Big data

Statistical / Numerical
computing

Types of Data Structures in Pandas

Data Structure	Dim ⁿ	Description
Series	1	1-D labeled homogenous size immutable
Dataframe	2	General 2-D labeled heterogenous size mutable
Panel	3	3-D labeled size mutable

Pandas Dataframe

A dataframe is a 2-D Structure
(rows & column) in which pandas
store the data

Features

- ⇒ Can perform arithmetic operation
- ⇒ Analogous to tabular structure
- ⇒ size mutable
- ⇒ labeled columns.

Column Label/ Header

Index Label

	0	1	2	3	4	
Label/ Header	Name	Age	Marks	Grade	Hobby	
0	S1	Joe	20	85.10	A	Swimming
1	S2	Nat	21	77.80	B	Reading
2	S3	Harry	19	91.54	A	Music
3	S4	Sam	20	88.78	A	Painting
4	S5	Monica	22	60.55	B	Dancing

Column Index

Row

Row Index

Column

Element/ Value/ Entry

Operations on Pandas DataFrame

Column

Add/delete /
select

Rows

Add/delete

Methods in Ponder

described